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TRANSPORT PHENOMENA I, CHE 311

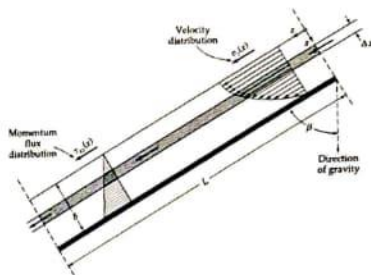
QUIZ 1

CLOSED BOOK, TIME: 15 MINUTES

The velocity profile of a liquid film flowing down an inclined flat plate is given by:

$$v_z = \frac{\rho g \cos \beta \delta^2}{2\mu} \left(1 - \frac{x^2}{\delta^2}\right)$$

- 1- Write down an equation for the z-component of the mass flux as a function of x.
- 2- Calculate the viscous tension on the gas-liquid and the solid-liquid interfaces.



All your answers must be in terms of the following constant parameters.

μ and ρ : viscosity and density of the liquid

δ : film thickness

β : angle of the flat plate relative to the direction of gravity

g : gravitational constant

Newton's law of viscosity: $|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$

BSL textbook notation: $\tau = -\mu(\nabla v + (\nabla v)^T)$

Something isn't right here $\rightarrow \rho v_x (\delta \tau)_z - \rho v_x (\delta \tau)_{z-1} + \rho v_x (\delta \tau)_z - \rho v_x (\delta \tau)_{z-1}$

$$\tau_{xz} = \begin{vmatrix} \tau_x \tau_x & \tau_x \tau_y & \tau_x \tau_z \\ \tau_y \tau_x & \tau_y \tau_y & \tau_y \tau_z \\ \tau_z \tau_x & \tau_z \tau_y & \tau_z \tau_z \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \tau_z \tau_x & 0 & 0 \end{vmatrix}$$

$$|\tau| = -\mu(|\nabla v| + |\nabla v|^T)$$

$$g \cos \beta$$

I was sick when you showed this in class, and didn't get to have this explained to me in lecture : I'll do better on the next quiz!

$\frac{0}{100}$